

Machine Learning for Aquatic Ecotoxicity

Prediction with Quantified Uncertainty

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Abstract

Aquatic ecosystems face increasing pressure from the thousands of industrial chemicals entering rivers and lakes annually, yet our ability to experimentally assess the risks of these compounds lags significantly behind their production. To protect these environments, regulators rely on toxicity data—specifically Lethal Concentration 50% (LC50) values—to determine safe thresholds. However, the global matrix of potential chemical-species interactions is overwhelmingly sparse, with over 99.5% of possible combinations untested. While machine learning techniques, particularly Factorization Machines (FM), have shown promise in “filling these gaps” by predicting missing toxicity values, they typically provide only point estimates. In a domain where safety is paramount, a single predicted number is insufficient; regulators and risk assessors require a measure of confidence to make informed decisions.

This article presents a robust solution to the sparsity problem: the Bayesian Factorization Machine (BFM). By re-framing the model parameters as probability distributions rather than fixed constants, the BFM not only achieves state-of-the-art predictive accuracy—matching

the noise floor of inter-test variability—but also provides a rigorous quantification of uncertainty. Crucially, the model decomposes this uncertainty into two distinct components: aleatoric uncertainty, reflecting the inherent noise and variability in biological testing, and epistemic uncertainty, reflecting the model’s lack of knowledge in unexplored areas of the data.

We demonstrate how this quantified uncertainty transforms downstream risk assessment workflows. First, we show how BFM predictions can be used to construct Species Sensitivity Distributions (SSDs) that include credible intervals and posterior distributions for hazardous concentration thresholds (HC5), moving beyond static curves to dynamic, uncertainty-aware risk profiles.

Furthermore, we explore the practical utility of epistemic uncertainty through a simulation of Active Learning. We illustrate how an "uncertainty subsampling" strategy can leverage the BFM’s outputs to intelligently prioritize future laboratory tests. By targeting experiments that maximize information gain, this approach can achieve target accuracy levels with significantly fewer samples than random testing, offering a pathway to more cost-effective and efficient environmental protection.

Dataset

The dataset used throughout this thesis is the ADORE dataset introduced by ?, which is a dataset designed for fostering the adoption of Machine Learning in ecotoxicology. The ecotoxicity values in ADORE come from the US EPA ECOTOX dataset[?], and ADORE further includes a large number of features of chemicals and species forming a low barrier to entry for data scientists from outside the domain, and reducing the risk of errors when procuring features. Multiple endpoints are available, such as Effective Concentration or No Effect Concentration, however out of the available endpoints, LC50 is chosen as the sole endpoint to focus on for all experiments, as it is consistently well-defined and has a significant

amount of data available. Future work could work on using data from multiple endpoints to improve model performance, however this would require significant data engineering work to standardise the reporting of other endpoints.

Each row of the dataset contains an LC50 observation, along with information relating to the chemical that was tested, the species that was exposed, and further additional information about the lab conditions for some observations. The high feature-richness of this dataset is what makes it an attractive choice for experimenting with Machine Learning, especially for practitioners outside of the ecotox domain.

The ADORE dataset includes a number of 'challenge' slices, where the data is limited to specific sub-species, however, in order to leverage as much of the data as possible, the processed but not divided dataset is used throughout this thesis. Specifically `ecotox_mortality_processed.csv` is joined with `ecotox_properties_with-oecd-function.csv` available to download at <https://gitlab.renkulab.io/mltox/adore>, where in-depth detail on the dataset processing can also be found.

Quantitative properties of the dataset

Table 1: Summary statistics of the ADORE dataset

Statistic	Value
Number of unique chemicals	3,295
Number of unique species	1267
Number of durations	4
Number of unique (species, chemical, duration) triplets observed	31,319
Total number of observations	70,670
Sparsity (species*chemical matrix)	99.5%

Table 1 shows summary statistics about the dataset, and elucidates the first major challenge of this dataset - **sparsity**. Even if the duration dimension is collapsed, only around 0.05% of the chemical-species pairs have observed LC50 values. This feature of the dataset informs further modelling choices, and highlights the inherent difficulty of the problem. Even filling this matrix is a major modelling challenge, and the amount of data is truly miniscule

when the size of the chemical universe is considered (around 350,000 chemicals are in active use in europe[?]).

The second major challenge is the **uneven distribution of observations** across the matrix, which is illustrated in Figure 1 from the point of view of chemicals, and species separately. The rank-frequency plots illustrate that most unique species and chemicals have just a handful of observations, while a few chemicals and species have been tested hundreds of times.

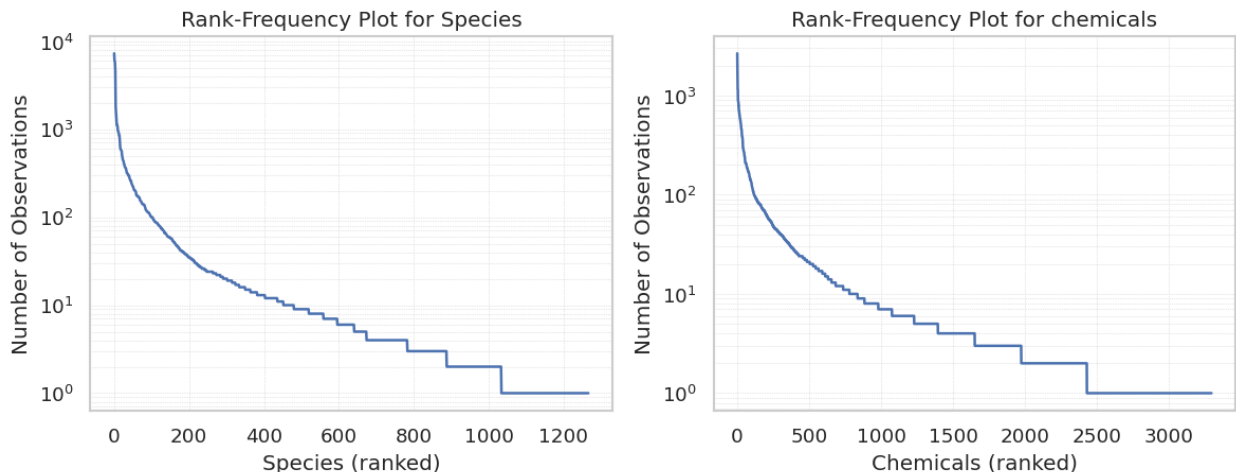


Figure 1: Plots showing the number of observations per species (left) and per chemical (right) ranked by number of observations.

The third challenge of the data is very high inter-test variability. Figure 2 shows the distribution of normalized standard deviations for chemical*species*durations where at least 10 observations are available. This shows that there is a significant portion of the dataset where the variance of repeated observations is very high, more than the mean or even double or triple the mean.

These three challenges are not unique to the ADORE data, but a common issue in ecotox datasets, and these qualities drive model choice and evaluation approaches in applying machine learning in this domain.

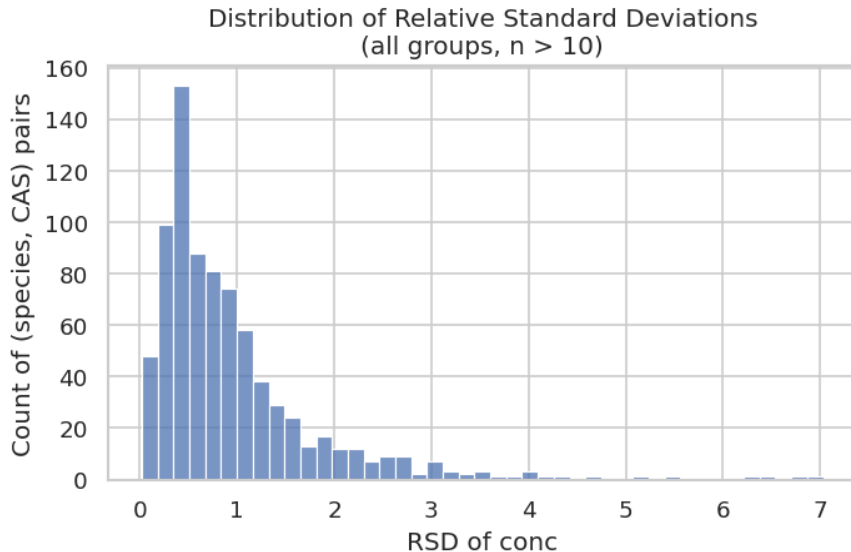


Figure 2: Distribution of RSD’s in the data grouped by species, chemical, duration

Methods

Hierarchical Bayesian Factorization Machine

We select the Factorization Machine (FM) as our foundational architecture due to its proven state-of-the-art performance on the ADORE dataset and its specific efficacy in handling high-dimensional, sparse ecotoxicological data. However, standard predictive implementations—such as basic Gaussian process regression or neural networks—typically rely on a homoscedastic noise assumption, learning a single global variance parameter. This approach is limiting in the context of ecotoxicity, as it conflates the inherent variability of well-studied, stable chemicals with that of poorly characterized or volatile ones, leading to poorly calibrated confidence intervals.

To address this limitation, we introduce the Hierarchical Bayesian Factorization Machine for Ecotox LC50 prediction. Adapted from the implementation by Rendle (2012), this model extends the classical framework by incorporating a per-chemical (group-specific) noise precision parameter regularized by a shared hierarchical prior. This formulation allows the model to: (1) learn chemical-specific aleatoric uncertainty, distinguishing between inherently noisy

and consistent data; (2) achieve shrinkage towards a global mean, allowing rare chemicals to borrow statistical strength from the population; and (3) explicitly decompose predictive uncertainty into irreducible aleatoric noise and reducible epistemic uncertainty.

Full Model Definition

Let the dataset consist of N observations. Each observation i is characterized by: $\mathbf{x}_i \in R^p$, a sparse feature vector encoding chemical identity, species identity, test duration, taxonomic family, taxonomic class, and numerical chemical properties; $y_i \in R$, the observed log-transformed toxicity value; and $c_i \in \{1, \dots, C\}$, the index of the chemical (group needed for noise term α_c) associated with observation i .

The response y_i is modeled as a noisy observation from a factorization machine:

$$y_i \mid \mathbf{x}_i, \theta, \alpha_{c_i} \sim \mathcal{N}(\hat{y}_i(\theta), \alpha_{c_i}^{-1}) \quad (1)$$

where $\alpha_c > 0$ is the *precision* (inverse variance) specific to chemical c , and the predictive mean $\hat{y}_i(\theta)$ is given by a second-order factorization machine:

$$\hat{y}_i(\theta) = w_0 + \sum_{j=1}^p w_j x_{ij} + \sum_{j=1}^p \sum_{j' > j}^p \langle \mathbf{v}_j, \mathbf{v}_{j'} \rangle x_{ij} x_{ij'} \quad (2)$$

where $w_0 \in R$ is the global bias (intercept), $\mathbf{w} = (w_1, \dots, w_p)^\top \in R^p$ are the linear (main effect) weights, $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_p)^\top \in R^{p \times k}$ is the matrix of latent factor vectors with each $\mathbf{v}_j \in R^k$, and $\langle \mathbf{v}_j, \mathbf{v}_{j'} \rangle = \sum_{f=1}^k v_{jf} v_{j'f}$ is the inner product capturing pairwise interactions.

Hierarchical Prior Structure.

The global bias is assigned a Gaussian prior:

$$w_0 \sim \mathcal{N}(\mu_0, \gamma_0^{-1}) \quad (3)$$

with $\mu_0 = 0$ and $\gamma_0 = 1$. The linear weights are modeled with a shared Normal-Gamma prior inducing adaptive shrinkage:

$$w_j \mid \mu_w, \lambda_w \sim \mathcal{N}(\mu_w, \lambda_w^{-1}) \quad \text{for } j = 1, \dots, p \quad (4)$$

$$\mu_w \mid \lambda_w, \mathbf{w} \sim \mathcal{N}(\bar{w}_{\text{post}}, (\gamma_0 + p\lambda_w)^{-1}) \quad (5)$$

$$\lambda_w \mid \mathbf{w}, \mu_w \sim \text{Gamma}\left(a_\lambda + \frac{p}{2}, b_\lambda + \frac{1}{2} \sum_{j=1}^p (w_j - \mu_w)^2\right) \quad (6)$$

with weakly informative hyperpriors $a_\lambda = b_\lambda = 1$. Analogously, the latent factors share a Normal-Gamma prior:

$$v_{jf} \mid \mu_v, \lambda_v \sim \mathcal{N}(\mu_v, \lambda_v^{-1}) \quad \text{for all } j, f \quad (7)$$

$$\mu_v \mid \lambda_v, \mathbf{V} \sim \mathcal{N}(\bar{v}_{\text{post}}, (\gamma_0 + pk\lambda_v)^{-1}) \quad (8)$$

$$\lambda_v \mid \mathbf{V}, \mu_v \sim \text{Gamma}\left(a_\lambda + \frac{pk}{2}, b_\lambda + \frac{1}{2} \sum_{j,f} (v_{jf} - \mu_v)^2\right) \quad (9)$$

Extension to Hierarchical Per-Chemical Precision.

Each chemical's precision is drawn from a shared Gamma prior:

$$\alpha_c \sim \text{Gamma}(a_0, b_0) \quad \text{for } c = 1, \dots, C \quad (10)$$

where $a_0 = b_0 = 1$ (an exponential prior with mean 1). Given the observations, the posterior for each α_c is conjugate:

$$\alpha_c \mid \{y_i, \hat{y}_i : c_i = c\} \sim \text{Gamma}\left(a_0 + \frac{N_c}{2}, b_0 + \frac{1}{2} \sum_{i:c_i=c} (y_i - \hat{y}_i)^2\right) \quad (11)$$

where $N_c = |\{i : c_i = c\}|$ is the number of observations for chemical c . Chemicals with many observations will have posteriors dominated by the data's sum of squared errors, while chemicals with few observations will have posteriors strongly influenced by the prior, achieving

partial pooling.

Inference via Gibbs Sampling

Posterior inference is performed using a Gibbs sampler, which iteratively samples each parameter from its full conditional distribution. For full details see `rendle` or the code.

The sampler is run for T total iterations, with the first T_{burn} samples discarded as burn-in.

Prediction and Uncertainty Quantification

Posterior Predictive Mean. For a new observation with features $\mathbf{x}^*$ belonging to chemical c^* , the posterior predictive mean is:

$$\hat{y}^* = \frac{1}{S} \sum_{s=1}^S \hat{y}^{(s)}(\mathbf{x}^*) \quad (12)$$

where $\hat{y}^{(s)}(\mathbf{x}^*)$ is the prediction from the s -th posterior sample and S is the number of post-burn-in samples.

Epistemic Uncertainty. Epistemic uncertainty reflects uncertainty in the model parameters due to limited training data:

$$\text{Var}_{\text{epistemic}}(\mathbf{x}^*) = \frac{1}{S} \sum_{s=1}^S \left(\hat{y}^{(s)}(\mathbf{x}^*) - \hat{y}^* \right)^2 \quad (13)$$

This uncertainty is reducible—it decreases as more data becomes available.

Aleatoric Uncertainty. Aleatoric uncertainty reflects inherent variability in the data-generating process:

$$\text{Var}_{\text{aleatoric}}(c^*) = \frac{1}{S} \sum_{s=1}^S \frac{1}{\alpha_{c^*}^{(s)}} \quad (14)$$

where $\alpha_{c^*}^{(s)}$ is the s -th posterior sample of the precision for chemical c^* .

Total Predictive Variance. The total predictive variance combines both sources:

$$\text{Var}_{\text{total}}(\mathbf{x}^*, c^*) = \text{Var}_{\text{epistemic}}(\mathbf{x}^*) + \text{Var}_{\text{aleatoric}}(c^*) \quad (15)$$

distribution at that rank.

Results

Predictive accuracy

For $k = 32$, $n_{\text{iter}} = 2000$, $n_{\text{burn}} = 50$ the RMSE is 0.71 (log mg/L) in a 5-fold cross-validation setup grouped by CAS, species and duration. This means the model will not see results for the same triplet at train and test time. The RMSE is state of the art performance on this view of ADORE (cf.,¹²), with comparisons to other methods evaluated on other datasets being difficult.

Figure 3 shows a correlation plot of the model predictions for species/chemical pairs at 48 hour exposure versus the average of the actual observations for those pairs. The figure shows a strong correlation and gives a visual impression of the calibration of the model.

A further inspection of the residuals in Figure 4 allows for a deeper understanding of the biases of the model. Panel 1 (leftmost) shows a scatterplot of the residuals against the predicted toxicity value. Aside from a few extremely outlying points, the shape of the plot is relatively uniform and suggests calibrated model behaviour. The second panel (centre) shows the residuals scattered against the measured observations, showing a negative diagonal slope, indicating regression to the mean happening (very high measured values tend to be under-predicted while very low values tend to be over-predicted).

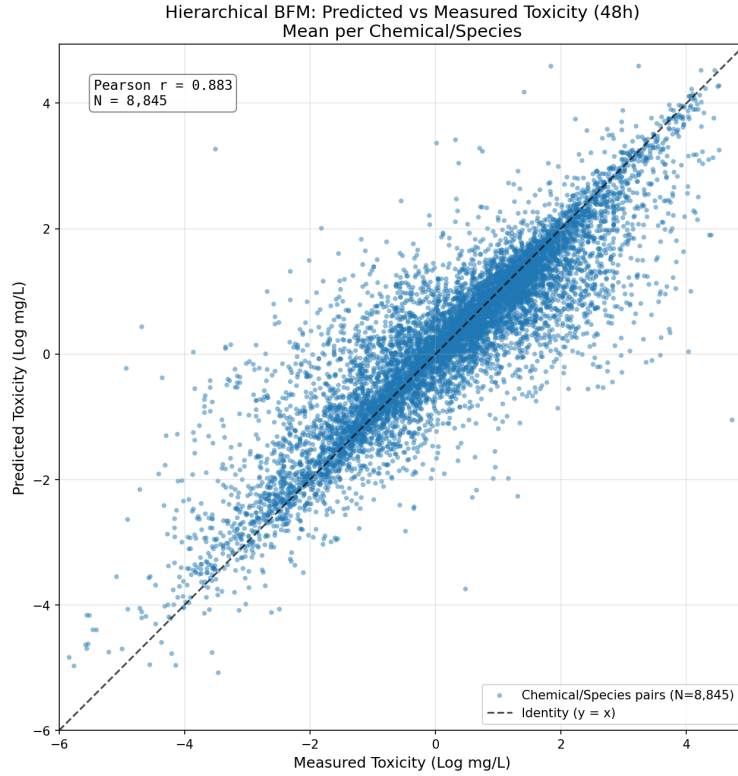


Figure 3: Each point on the plot is the mean measured $\log(\text{mg/L})$ toxicity (average where multiple measurements available) vs. the mean predicted $\log(\text{mg/L})$ toxicity for a unique chemical–species pair at 48 hours.

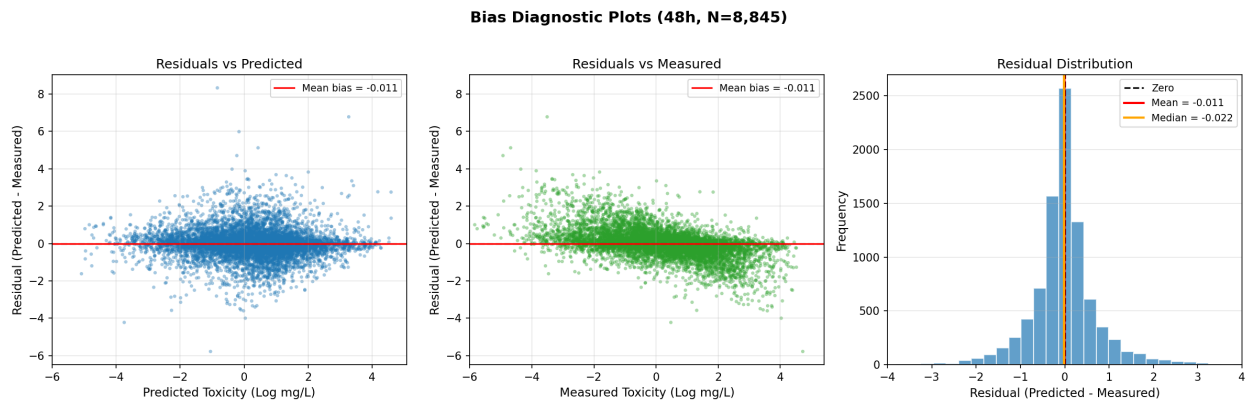


Figure 4: Residual diagnostic plots showing (a) residuals vs predicted values, (b) residuals vs measured values revealing regression to the mean, and (c) residual distribution histogram.

Uncertainty calibration

In the previous section descriptive statistics of the model’s predicted LC50 values were shown, without giving consideration to the uncertainty quantification metrics the model provides. This section aims to provide an evaluation of those uncertainty predictions.

As a reminder, the epistemic uncertainty relates to the variance between the predicted means of the n samples that make up the full bayesian model. This can also be called the modelling uncertainty, following the intuition that it represents how different models can be fitted to the data giving rise to different predictions. Intuitively, the more datapoints become available, the less modelling uncertainty we would expect to be - the more data there is, while keeping the expressiveness of the model constant, the less room there is for discrepancy between fits.

The aleatoric uncertainty is a term predicted directly by the model itself. Intuitively, it is meant to capture the natural variability between measurements, capturing effects such as biological variability, inter-test variability etc. In the current implementation it is captured at the per-chemical level, a compromise discussed in the methods section. In practice this term will capture any variability in the toxicity endpoint that the model cannot explain using other features.

Total variance can be considered the epistemic + aleatoric variance together, although depending on the specific application it can be better to use one or the other rather than the combination. Around 15% of total variance is epistemic, the rest is predicted by the model to be aleatoric. An intuitive way to interpret this is that the model is relatively confident in predicting the mean toxicity for a species/chemical, but is aware of a large spread in individual measurements.

Table 2 shows how the aleatoric uncertainty predictions relate to the empirical spread among repeat observations. The predicted uncertainty is roughly in the same range as the empirical, though with a tendency to overestimate, with more replicates reading to more agreement between the model and the empirical spread. This could be an artifact of the

dataset, or expected behaviour of the model learning the SD terms that satisfy the largest number of loss points, with the chemicals with hundreds of repeated measurements having more influence on the training loss.

Table 2: For triplets with repeated measurements, the empirical standard deviation is compared to the predicted SD (aleatoric uncertainty).

Replicates	N groups	Mean Ratio	Median Ratio	Mean Pred SD	Mean Emp SD
5–9	1,803	3.605	1.753	0.567	0.387
10–19	658	2.534	1.635	0.521	0.381
20–49	227	1.877	1.436	0.516	0.400
50–99	30	1.532	1.153	0.537	0.463
100+	6	1.014	0.968	0.537	0.558

To examine whether the model’s uncertainty estimates respond appropriately to data availability, we compared out-of-fold epistemic and aleatoric standard deviations against the total number of observations per chemical (Figure 6). Epistemic uncertainty showed a strong negative association with observation count (Spearman $\rho = 0.649$), consistent with the expectation that additional data constrains the chemical’s latent factors and reduces parametric uncertainty.

The aleatoric component showed an even stronger negative association (Spearman $\rho = 0.824$). This parameter, α_c , is estimated per chemical and represents the residual variance not explained by the model’s mean function — encompassing measurement noise, inter-species and inter-duration variability beyond what the latent factor structure captures, and any other chemical-specific model misfit. The strong dependence on observation count likely reflects a combination of factors: (i) the Gamma(1, 1) prior on α_c is relatively diffuse, so data-poor chemicals are shrunk toward a prior mean variance of 1, while data-rich chemicals are dominated by their empirical residual variance; (ii) well-studied chemicals may be associated with more standardised testing protocols and better-characterised dose-response relationships; and (iii) chemicals with extensive testing histories tend to span more species and durations, but the model has correspondingly more data from which to learn their latent

factors, resulting in tighter residuals.

The aleatoric parameter should not be interpreted as irreducible measurement noise in isolation: because α_c is shared across all species and durations for a given chemical, it absorbs any residual variance the model’s rank-32 factorisation does not capture. The total predictive uncertainty for a given (chemical, species, duration) triplet is the sum of the epistemic and aleatoric components.

To give a visual idea of what the model predictions with uncertainty look like in practice, figure 5 shows examples of the model prediction for five selected chemicals, and the species for which they were tested in the dataset, to compare side-by-side test results with predictions. At the same time the aleatoric uncertainty bar shows the spread of observations around the mean predicted by the model, while the dark blue bars represent the model uncertainty around the mean.

SSD’s

Species Sensitivity Distributions (SSDs) are tool used in hazard assessment to capture the relationship between the concentration of a specific chemical and the percentage of species in an ecosystem that will be affected by it. SSDs are typically constructed by fitting a cumulative distribution function to observed toxicity data for a given chemical. The accuracy of this method is reliant on using a representative sample of species from the ecosystem.

A novel approach to SSDs is introduced in Posthuma et al.², where a toxicity-prediction model is fitted to an ecotox dataset, and then is used to predict toxicity values for all species/chemical combinations among those present in the data. Such a model implicitly performs read-across on both the species and chemical level. Since the model provides a prediction for every species in the dataset, curve-fitting is no longer necessary, and a scatterplot of the predictions ordered by concentration yields the familiar CDF shape.

The model used in the current work extends that of,² by also including selected chemical properties to improve predictive accuracy, and more importantly by using bayesian modelling

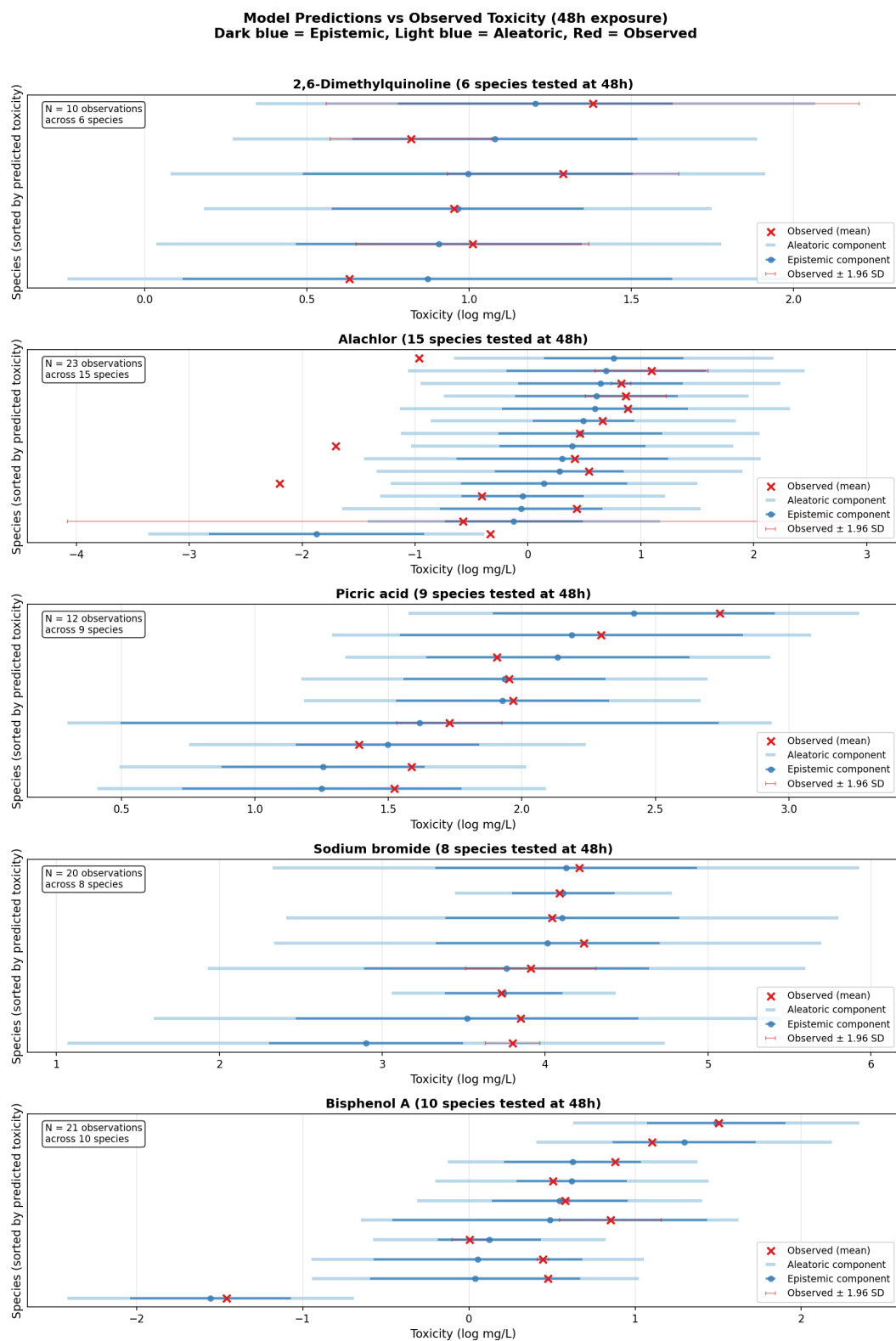


Figure 5: Predicted vs observed toxicity with decomposed uncertainty for five randomly selected chemicals at 48 h exposure. For each chemical, species are sorted vertically by predicted toxicity. Blue dots show the model's mean prediction; dark blue bars indicate ± 1.96 epistemic SD (posterior parametric uncertainty), and light blue bars extend to ± 1.96 total SD (adding the per-chemical aleatoric component). Red crosses show the mean observed toxicity, with red bars indicating ± 1.96 SD of replicate measurements where available.

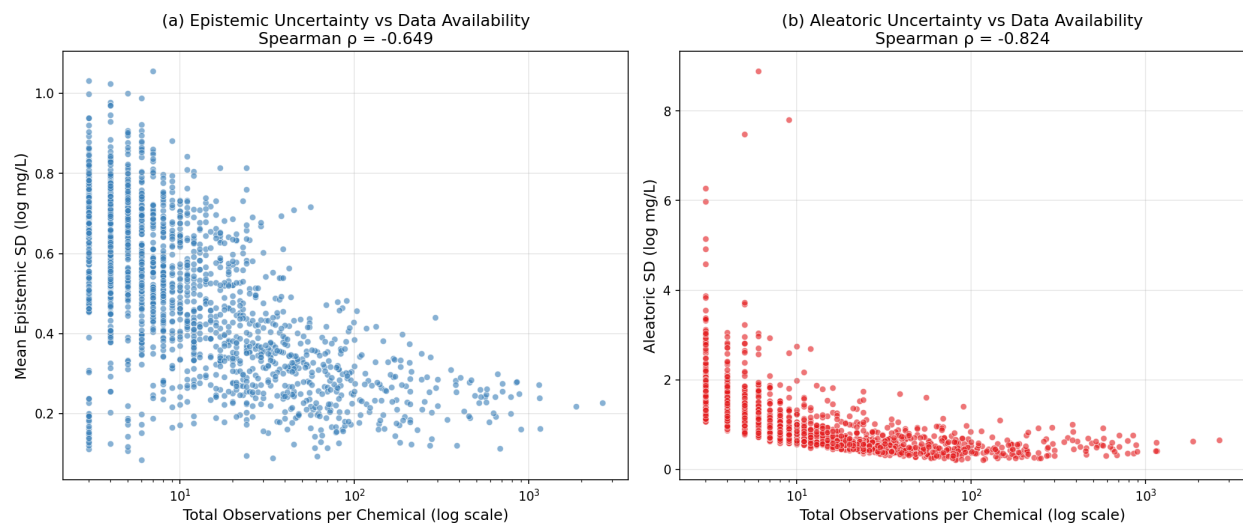


Figure 6: Relationship between data availability and predicted uncertainty. Each point represents one chemical at 48 h exposure duration. The x-axis shows the total number of observations (across all durations) available for that chemical in the training set. Chemicals with fewer than three total observations were excluded to avoid cold-start artifacts in the cross-validated uncertainty estimates ($N = 660$ excluded; $N = 1,722$ retained). (a) Mean epistemic standard deviation (SD), computed as the average across out-of-fold posterior predictive variance for each chemical’s 48 h observations. (b) Mean aleatoric SD, derived from the per-chemical precision parameter α_c . Both uncertainty components are expressed in log mg/L units.

to provide uncertainty quantification to the predictions. The current section demonstrates how the uncertainty around the predictions can be propagated to SSDs and HC values.

7a shows a traditional SSD for Atrazine at 48 hour exposure, fitted to the observations in the ADORE dataset. 7b shows the analogous novel format. 8 shows the novel and traditional SSDs for Atrazine on the same panel (8a, and next to it shows the same comparison for chlorfenprop methyl, which was selected for it's small number of test results in the dataset, and shows a much higher discrepancy from the fitted SSD (8b).

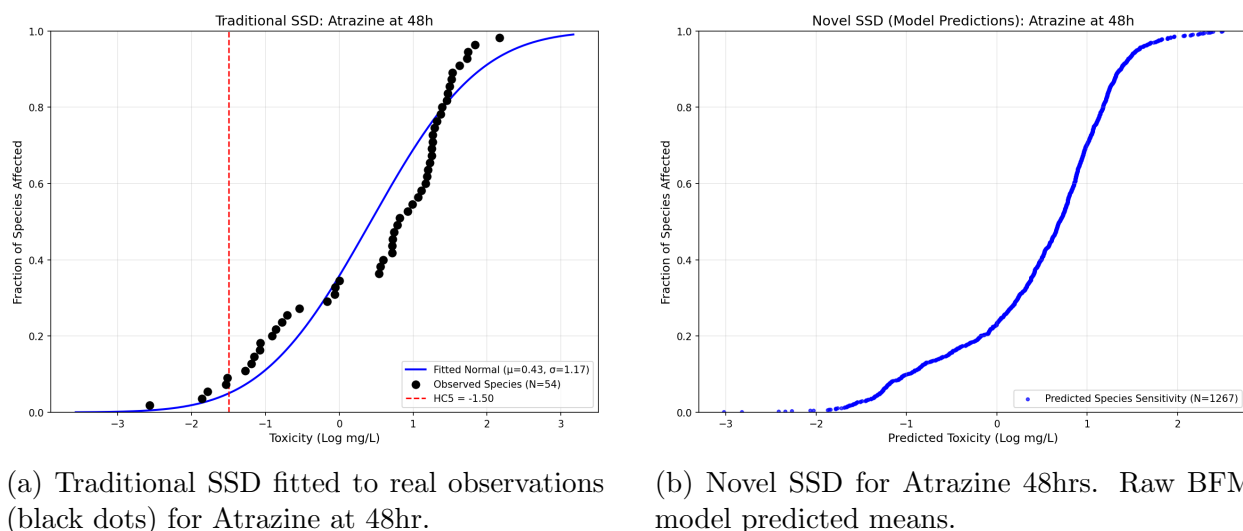


Figure 7: Comparison of Traditional vs. Novel SSD approaches for Atrazine at 48hr exposure.

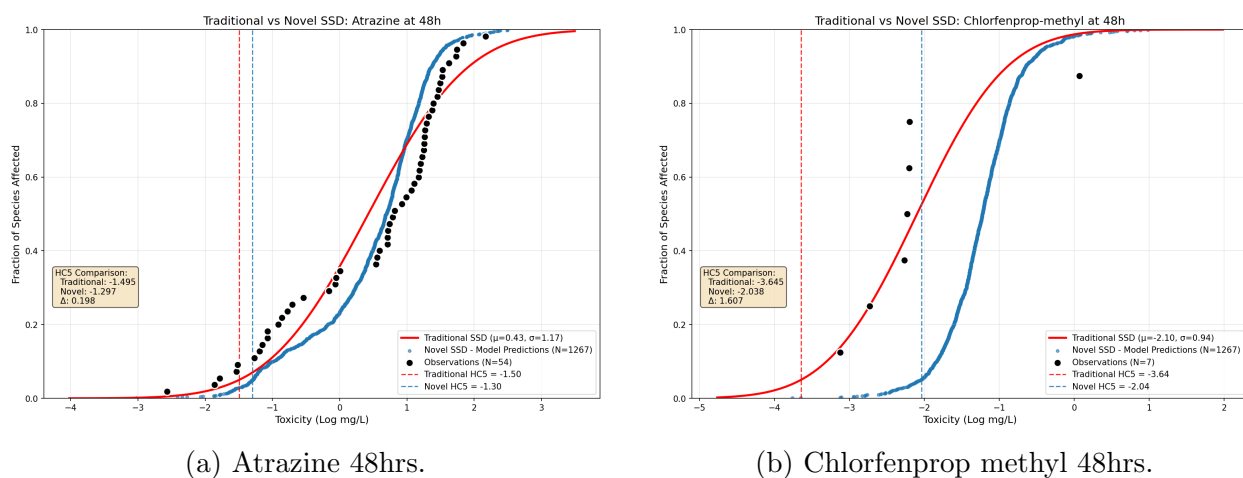


Figure 8: Side-by-side comparison of Novel vs. Traditional SSDs for two different compounds.

Species Sensitivity Distributions with uncertainty

The Bayesian Factorization Machine Model predicts a variance term for each chemical in addition to the mean toxicity prediction for each chemical/species pair, which is intended to capture the biological and experimental variance based on the dataset on a per-chemical granularity. Additionally for each prediction, the variance of the predicted means from each model in the sample shows the modelling uncertainty.

The plots below show the Novel SSD plot for Atrazine and Chlorfenprop-methyl where for each plotted prediction, its two associated uncertainty metrics are shown. Once again Atrazine is selected as a commonly tested chemical, while chlorfenprop methyl is a sparse chemical, having measurements for only 7 unique species (vs 144 for Atrazine).

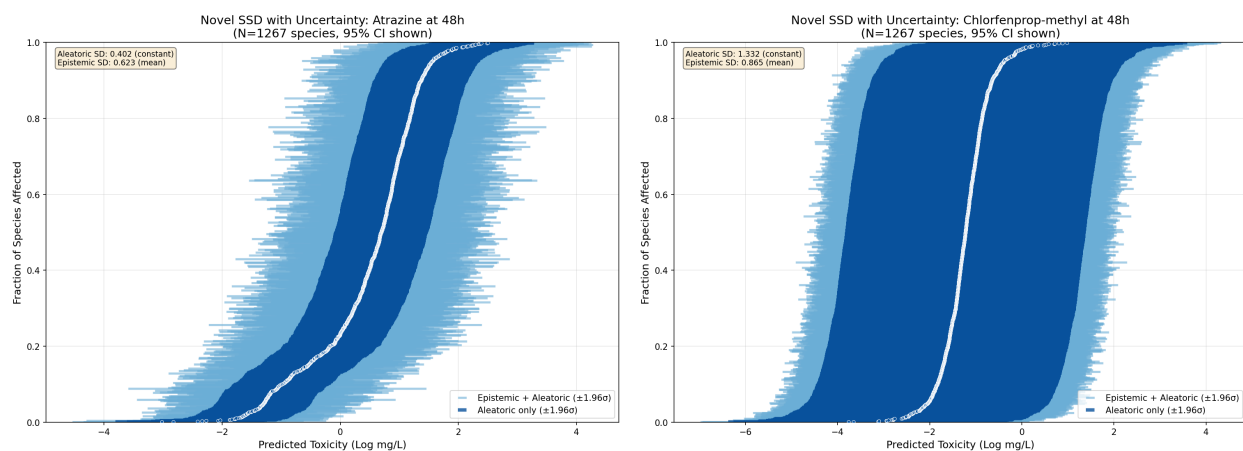


Figure 9: Novel scatter-SSDs for Atrazine (left) and Chlorfenprop (right) with uncertainty bars added for the uncertainty around model predictions for each point. Dark blue is the aleatoric uncertainty, which is constant for each chemical and predicted by the model. Light blue shows the additional epistemic uncertainty, which is different for each species and chemical.

The comparison side-by-side shows how a less tested compound has much higher uncertainty around the predictions, and it is noteworthy the difference in aleatoric (constant, predicted by the model) uncertainty is much greater (around 300%) than the difference in epistemic (modelling) uncertainty (around 40% greater).

Adding the uncertainty bars to the SSD plots in this way is a nice way to visualise the uncertainty around each point, but it should **not be interpreted as the uncertainty of**

the SSD, it shows the uncertainty around each point.

To propagate uncertainty from the predictions to the SSD, we instead construct the SSD directly from the MCMC posterior samples retained during model training. For each of the $S = 1950$ post-burn-in Gibbs samples, we compute the deterministic toxicity prediction $\hat{y}_i$ for every species i using the full factorisation machine equation evaluated at $\theta^{(s)}$. Each set of predictions is then sorted to yield a complete SSD curve, producing an ensemble of S internally consistent SSDs. Because each curve is derived from a single coherent parameter draw, the correlated epistemic structure across species is preserved: the entire SSD shifts left or right as the shared parameters vary, rather than individual points scattering independently. Credible intervals for any functional of the SSD, such as the HC5, can then obtained directly from the empirical distribution of that quantity across the ensemble of posterior SSDs. Figure 10 shows these ensemble-SSDs plotted as semi-transparent lines, giving a visual impression of the uncertainty around the SSDs stemming from the modelling process. The SSDs for Atrazine are much closer together, appearing almost as a single solid line, while those for Chlorfenprop-Methyl show a larger spread.

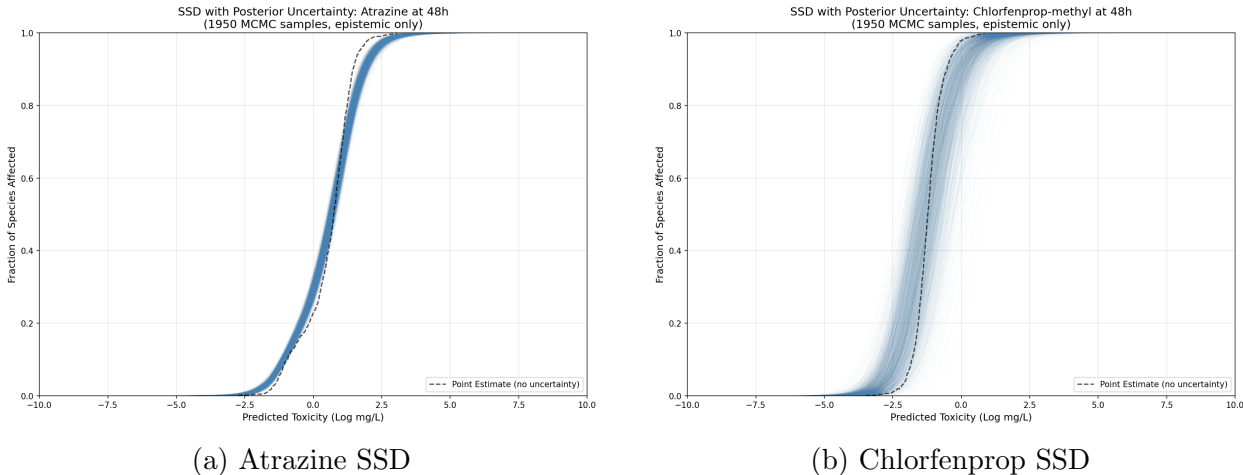


Figure 10: Comparison of SSD results for Atrazine and Chlorfenprop

These SSDs capture the uncertainty around the modelling process, allowing for uncertainty-aware derivation of Hazardous Concentrations. Figure 11 shows HC20 values along with their 95% confidence intervals and HC20 values derived from lognormal curve-fitting for compar-

ison. Many of the traditionally fitted HC20 values fall outside of the confidence interval of the BFM predicted ones, and tend to be more liberal. The BFM tends to predict more conservative (i.e. more protective, lower) HC20 values. Figure 12 is a correlation plot for all chemicals which had enough (5) measurements to construct a curve-fitted SSD, of the HC20 values. The plot shows a high correlation, while also showing the systemic bias of the BFM predictions to predict lower HC20 values.

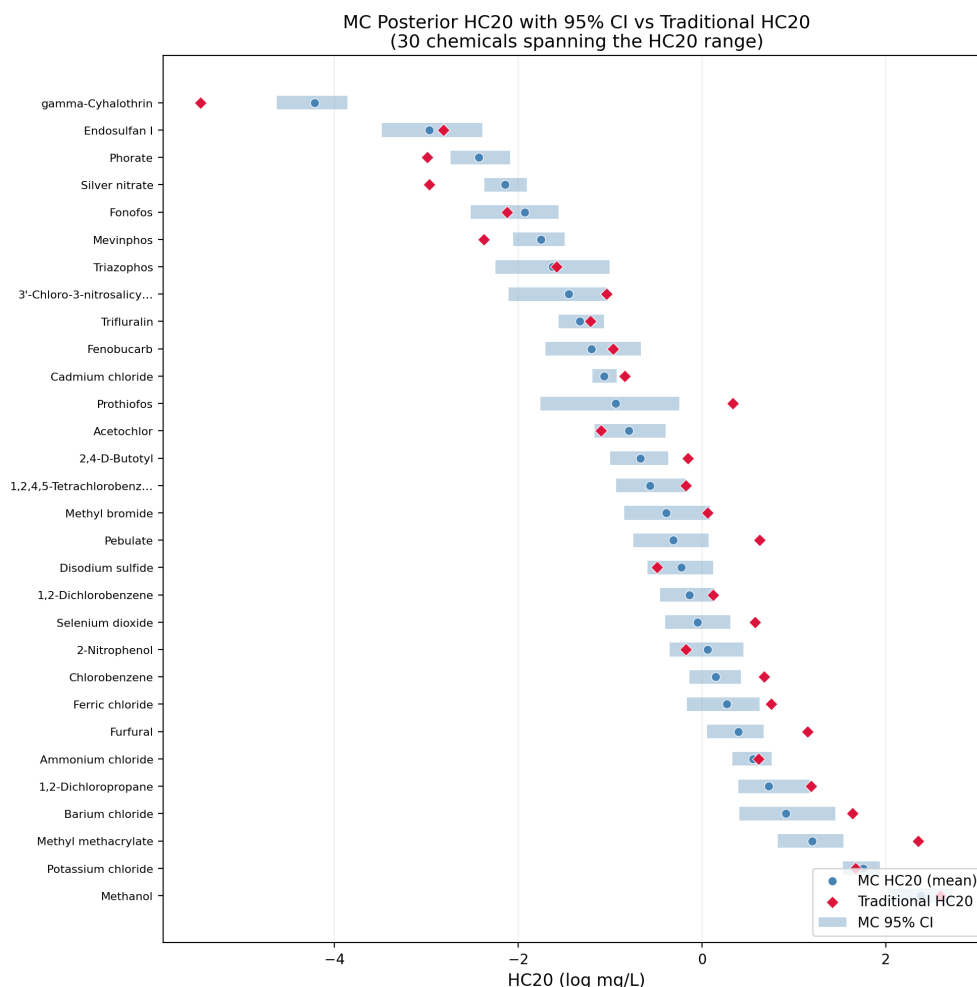


Figure 11: A plot comparing HC20 values stemming from the BFM model predictions and HC20 values derived from lognormal curves fitted to true observations (red).

Conclusion

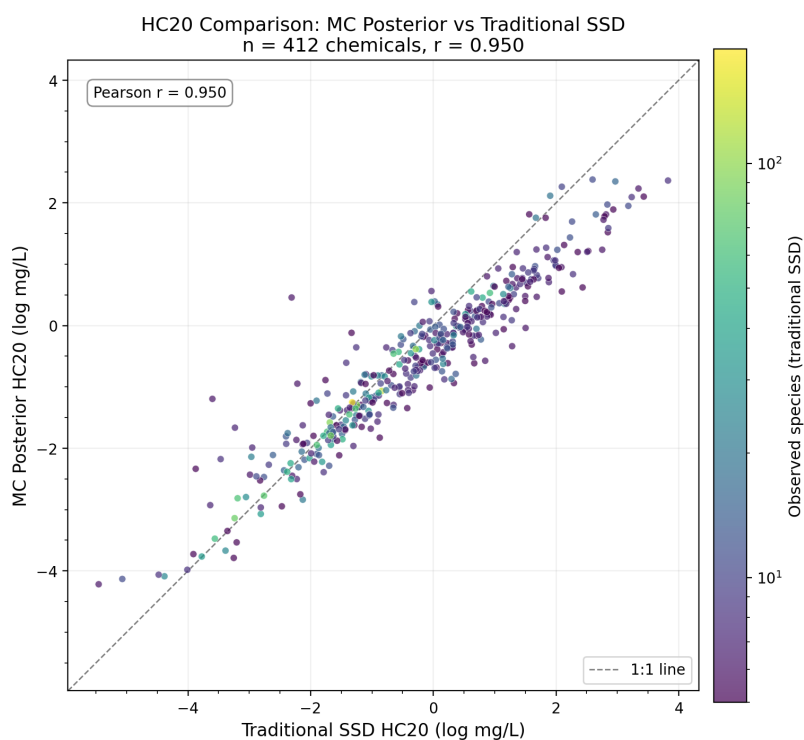


Figure 12: Correlation plot of lognormal fit derived HC20 versus HC20 derived from model predictions. Only for chemicals which had at least 5 observations for curve fitting validity. Coloured by number of observations per chemical.

Limitations

- no per-species uncertainty measurement
- little use of chemical and species information, making extending the model to untested chemicals impossible
-

Acknowledgement

References

- (1) Viljanen, M.; Minnema, J.; Wassenaar, P.; Rorije, E.; Peijnenburg, W. What is the ecotoxicity of a given chemical for a given aquatic species? Predicting interactions between species and chemicals using recommender system techniques. *SAR and QSAR in Environmental Research* **2023**, *34*, 765–788, PMID: 37670728.
- (2) Posthuma, L.; Price, T.; Viljanen, M. Improving the Ecotoxicological Hazard Assessment of Chemicals by Pairwise Learning. *Environmental Science & Technology* **2025**, *59*, 16250–16260, PMID: 40744438.